

EventTrace: Verifying Ordered Semantic Events for Reinforcement Fine-Tuning of Vision-Language Navigation

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Abstract

001 Reinforcement fine-tuning of vision-language navigation
002 (VLN) policies commonly relies on destination
003 success or similarity to a reference route. These signals
004 may leave an instruction's intermediate requirements
005 unresolved: reaching the endpoint does not establish
006 that the agent entered the requested room or
007 passed the specified landmark. We study an online reward
008 that represents an instruction as an ordered list
009 of semantic events and verifies each event against the
010 observations before and after an executed action. A
011 frozen vision-language model supplies a three-way verdict
012 (complete, incomplete, uncertain), while motion
013 and stopping checks rule out visually plausible but un-
014 executed events. An event is rewarded at most once
015 and the total process reward is bounded below a success
016 bonus. We implement this reward in a multi-turn
017 ActiveVLN/GRPO training pipeline and report a controlled
018 feasibility experiment on eight R2R training
019 episodes. Across two optimization steps per arm, event
020 and destination-only training each attain 13 successes
021 in 16 sampled rollouts. On a fixed exploratory 16-
022 episode val-unseen subset, the SFT, destination-only,
023 and event checkpoints each succeed in 6 episodes. A
024 separate blind review of 49 selected event transitions
025 finds that only 7 of 13 verifier completion predictions
026 agree with the review labels. The study demonstrates
027 reward integration but does not demonstrate a navigation
028 gain or reliable event verification.

029 1. Introduction

030 Vision-language navigation asks an embodied agent to
031 follow a natural-language route through visual obser-
032 vations. The Room-to-Room (R2R) task introduced
033 instruction-guided navigation in real indoor environ-
034 ments [1]; VLN-CE moves this task into continuous
035 control [5]. Success rate (SR) and success weighted by

path length (SPL) capture destination attainment, yet
a successful stop can coexist with skipped intermedi-
ate instructions. Instruction fidelity work made this
gap explicit [4].

Recent reinforcement fine-tuning (RFT) approaches
give visual-language policies online or trajectory-level
feedback. ActiveVLN performs multi-turn exploration
with group-relative policy optimization (GRPO) [8,
10]. VLN-R1 uses time-decayed action rewards [7];
Nav-R1 combines format, understanding, and naviga-
tion rewards [6]; SeeNav-Agent proposes step-level ver-
ifiable process rewards in another navigation bench-
mark [9]. Our question is narrower: can a naviga-
tion instruction's ordered, physical events provide a
bounded reward during closed-loop R2R-CE rollouts,
without matching the policy's actions to a reference
path?

We introduce EventTrace, a modular reward on top
of an existing multi-turn policy. The reward uses a
short event list, a frozen visual verifier with an absten-
tion state, deterministic checks of motion or stopping,
and a once-only order-preserving tracker. The actor
sees only the original egocentric observations and in-
struction; simulator displacement is available only to
the reward. Figure 1 illustrates the distinction between
seeing a bathroom and actually entering it.

The present manuscript reports a feasibility pilot
and a small held-out probe rather than a claim of im-
proved navigation. We make the implementation, mea-
sured counts, and limitations explicit so that a larger
study can test the central hypothesis without confusing
a changed reward scale with improved SR.

2. Related Work

Path and instruction fidelity. SR and SPL reward the
destination and efficiency, whereas nDTW and cover-
age weighted by length score (CLS) compare trajecto-
ries with a reference route [3, 4]. Reference trajecto-
ries can be useful supervision but do not directly an-

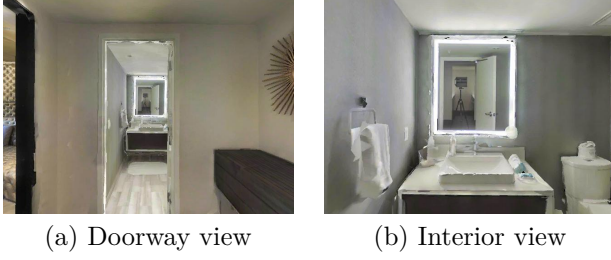


Figure 1. Real simulator frames from a diagnostic replay of R2R training episode 2889. Visibility alone does not complete the event “enter bathroom.” The verifier uses before/after views and executed motion; the ordered tracker later checks “stop near sink.” These frames illustrate the mechanism and are not an independent accuracy test.

074 swer whether a distinct visual event actually occurred
 075 when multiple valid routes exist. Our reward targets
 076 event completion during execution; route metrics re-
 077 main complementary evaluation measures.

078 RL for VLN. ActiveVLN is our implementation base
 079 because its environment can return feedback after each
 080 multi-action turn [10]. VLN-R1 and Nav-R1 show
 081 other ways to shape VLN rewards [6, 7]. SeeNav-
 082 Agent is especially relevant in its emphasis on step-
 083 level verification [9]. EventTrace differs in its ordered
 084 event state, explicit uncertainty outcome, and physi-
 085 cal gates against rewarding a landmark seen without
 086 the requested transition. Our pilot still applies GRPO
 087 to trajectory-level returns; it does not claim SeeNav-
 088 Agent-style step-level credit assignment.

089 3. Method

090 3.1. Task and event representation

091 An instruction x yields a list $E(x) = (e_1, \dots, e_K)$,
 092 where e_k contains a verb type, target phrase, and
 093 source span. Supported pilot events include enter,
 094 pass, descend, and stop_near. We first generate can-
 095 didate events using a frozen vision-language model
 096 (VLM), check the event verb and target against the
 097 instruction, and curate the eight lists used in the pilot.
 098 The measured pilot therefore tests reward integration
 099 given curated events; it does not validate automatic
 100 instruction parsing.

101 The parser is constrained to at most three events and
 102 the source span must appear in the original instruction.
 103 In the curated pilot, each of the eight episodes contains
 104 a stopping event; there are also three enter, three pass,
 105 and three descend events, for 17 in total. These counts
 106 describe the narrow pilot distribution, not the preva-
 107 lence of event types in R2R. Table 1 shows the deter-

Table 1. Motion gates in the pilot. The VLM must also verify that the image pair supports the specified target.

Event	Count	Required gate
enter	3	Forward action; $\ \Delta p\ \geq 0.10$ m
pass	3	Forward action; $\ \Delta p\ \geq 0.10$ m
descend	3	Vertical change $\Delta y \leq -0.08$ m
stop_near	8	Executed STOP action

ministic portion of the verifier. Each gate can reject
 completion before a VLM call, but passing a gate is
 never sufficient to award a point.

At turn t , the policy receives the instruction, action
 history, and current egocentric image I_t , and emits up
 to three low-level actions. The simulator executes these
 actions and returns I_{t+1} , pose change Δp_t , and task
 metrics. Only the current pending event e_{k_t} is queried.
 The frozen verifier produces

$$v_t = V(e_{k_t}, I_t, I_{t+1}, a_t, \Delta p_t) \in \{Y, N, U\}, \quad (1)$$

where U denotes insufficient evidence. A view of a
 landmark is not by itself a completion verdict. For
 crossing and passing, a forward displacement check is
 required; descending requires a negative vertical dis-
 placement; stopping near a target requires an executed
 STOP. The VLM inspects whether the image pair sup-
 ports the specified target and transition. These are
 deliberately conservative gates, not a learned proof of
 physical state.

3.2. Ordered, bounded process reward

Let $k_t \in \{1, \dots, K+1\}$ index the first unfinished event.
 The event fires only when its verifier says Y. Then

$$c_t = \mathbb{I}[k_t \leq K \wedge v_t = Y], \quad (2)$$

$$k_{t+1} = k_t + c_t, \quad r_t^{\text{sem}} = \frac{\lambda}{K} c_t. \quad (3)$$

A N or U verdict leaves k_t unchanged. The tracker
 prevents repeated reward for the same landmark, en-
 forces event order, and guarantees $\sum_t r_t^{\text{sem}} \leq \lambda$. Be-
 cause only the pending event is checked, at most one
 event fires per turn even if the agent performs several
 actions. This avoids ambiguous credit when a model
 emits an action batch but can delay recognition of ad-
 jacent events. The tracker retains a confirmed event
 after later image changes, so a completed event cannot
 be unconfirmed by viewpoint drift.

In the pilot, the base destination term is the
 weighted-success reward already used by ActiveVLN,
 with nDTW weight zero. We add a fixed success floor

b to both arms:

$$R(\tau) = \sum_t r_t^{\text{sem}} + S(\tau) \left[b + \alpha \left(1 - \min \left(\frac{d_T}{3}, 1 \right) \right) \right], \quad (4)$$

where $S(\tau)$ is simulator success after STOP, d_T is terminal distance in meters, $\alpha = 15$, $b = 2$, and $\lambda \in \{0, 1\}$. Hence any success receives at least 2 while a failure receives at most 1 semantic point. This priority holds for this nonnegative reward configuration; it does not imply that the verifier is correct. The bound also makes failure reward interpretable: partial event completion can distinguish two unsuccessful attempts without allowing a failed route to outrank a successful one solely through event bonuses. The STOP event is distinct from simulator success: it demands visual proximity to its instruction target, while $S(\tau)$ uses the simulator’s goal region. Disagreement between these signals is precisely a case the proposed reward aims to expose.

For each training episode, we sample two trajectories and normalize their returns within the group for GRPO [8]. Event rewards are emitted after each turn, but the current trainer sums them before constructing a trajectory advantage. Consequently, the implementation changes which trajectories are preferred, not which individual turn receives the policy gradient. Turn-level event timing is available in the log for future credit-assignment work.

3.3. Verifier failure policy

The verifier is a frozen Qwen3-VL-8B-Instruct model. A service timeout, malformed response, or uncertain decision yields zero event reward; it does not end the episode. The pilot requires no reference action prefix. Simulator pose is used only for reward verification, never added to the actor observation. The fixed event manifest is matched to both episode ID and exact instruction text to avoid assigning another episode’s event list. The verifier sees two RGB frames resized to fit within 448×448 pixels, the event type and target, its source phrase, the executed action text, translation magnitude, and vertical displacement. Its output is a deterministic one-letter Y/N/U response; generated explanations are not treated as confidence scores. This makes the reward externally auditable through a saved per-turn verdict, although a forced-choice label is still fallible. The HTTP verifier is separated from the simulator so it can be frozen and logged independently of policy updates.

4. Experimental Setup

Environment and policy. We use the R2R train split in VLN-CE, rendered from Matterport3D scans [2],

the ActiveVLN implementation, and an SFT-initialized Qwen2.5-VL-3B policy. Matterport3D data and derived imagery are subject to its [academic terms of use](#). Each turn contains at most three actions; the limit is 12 turns and 36 simulator steps. Actor sampling uses temperature 1.2 and two rollouts per item. GRPO trains on two A800 GPUs with learning rate 10^{-6} , a batch of four episodes, and two optimizer steps. An independent A800 runs the frozen verifier and simulator services. The same eight curated train episodes, SFT checkpoint, optimization configuration, and destination reward are used in both arms. The destination-only control sets $\lambda = 0$; the event arm sets $\lambda = 1$.

Diagnostics. We first replay the eight episodes with expert actions solely to inspect whether the verifier can identify expected events. These replays are not training rollouts or a held-out annotated test set. Then both training arms start from the same SFT checkpoint. We retain raw rollout JSONL, console logs, TensorBoard scalars, and optimizer checkpoints. We report success counts and reward components from those artifacts. The initial eight sampled action sequences match exactly across arms, enabling a direct first-step reward check; later actions can diverge after parameter updates.

Metrics and held-out probe. SR is the fraction of sampled trajectories satisfying the simulator goal condition; SPL also accounts for path efficiency. Event progress is confirmed events divided by events in that episode’s list. A verifier U count measures abstention, not correctness. Mean return uses each arm’s own reward definition and therefore cannot be interpreted as a cross-arm performance metric. For an initial held-out probe, we select 16 fixed R2R val-unseen episodes across 11 scenes by cycling through scenes in sorted order. The SFT model and both step-2 checkpoints run the same episodes, prompt and ActiveVLN evaluation protocol, with one generation per episode, temperature 0.2, top- p 0.8, a 12-turn limit, and a maximum of 76,800 image pixels. This fixed small subset is exploratory and is not the full official val-unseen evaluation. The evaluator uses the same environment-specific OpenAI-compatible request and image-resize shim for all three checkpoints.

Blind verifier review. Separately, we collect shortest-path follower trajectories from 16 different val-unseen episodes, of which 12 yield at least one automatically parsed event. Before querying the verifier, one AI-assisted reviewer inspects RGB image pairs, executed actions, and motion and fixes 49 three-way completion

Table 2. Observed pilot results on eight training episodes. Each arm ran two optimizer steps and produced 16 sampled rollouts. The event arm’s larger mean return is expected from the added reward and does not indicate better navigation.

Metric	Destination only	EventTrace
Successful rollouts	13/16	13/16
Mean trajectory return	9.325	10.455
Mean event progress	–	0.656
Confirmed event hits	–	23
Uncertain verifier turns	–	33
Verifier service errors	–	0

Table 3. Pilot outcomes separated by optimization step. Step 1 uses identical sampled action sequences in both arms, so its return difference of 0.6875 equals the mean semantic event credit. Step 2 follows different policy updates and cannot be paired causally.

Metric	Step 1		Step 2	
	Control	Event	Control	Event
SR (of 8)	7	7	6	6
Mean return	10.683	11.370	7.967	9.540
Event progress	–	0.688	–	0.625

labels across those 12 episodes. The sample includes apparent transitions, controls, and final stops. Labels are independent of verifier predictions, but there is no second reviewer or human ground-truth adjudication, and the selected class proportions are not representative of deployment traffic. We therefore treat this as a failure analysis, not a population accuracy estimate.

5. Results and Analysis

Reward reaches optimization. In the first optimizer step, both arms sample the same eight action sequences and attain seven successes. For every matched trajectory, the difference in total return equals its semantic event score, including a failed episode 7110 trajectory that receives 0.5 for completing “descend stairs” but never reaches the destination. The modified return is consumed by the training reward manager, and both arms save step-2 checkpoints after nonzero policy updates. This verifies the wiring of Eq. 4; it is not evidence that event reward improves the policy.

What the verifier recognizes. In diagnostic expert replays, four of eight episodes receive full event credit, two receive partial credit, and two receive none; the tracker confirms 11 of 17 curated events. For episode 2889, seeing the sink from outside the bathroom produces uncertain verdicts before entrance, while later

motion and view changes confirm enter. A rotation followed by premature STOP yields zero semantic credit in a negative replay. There are also false negatives: one shallow stair descent is rejected by the vertical-motion threshold, and one landmark is never confirmed. Because the events were curated using these episodes and the replays lack independent annotations, 11/17 must not be reported as verifier recall or accuracy.

Navigation outcome. Both arms have 13/16 successes after the two-step pilot (Table 2). The samples are correlated within eight training episodes, and each arm has only one run. The experiment supports feasibility of online event reward, not a claim of generalization or superior SR.

Exploratory val-unseen result. Table 5 reports the fixed 16-episode subset. All three checkpoints achieve 6/16 SR. EventTrace exceeds destination-only training by just 0.0009 SPL; the event checkpoint succeeds on two episodes where the destination-only checkpoint fails and fails on two episodes where it succeeds. The paired SR difference is zero, with a 95% episode-bootstrap interval of $[-0.25, 0.25]$. The paired SPL difference is 0.0009, with interval $[-0.238, 0.226]$. These wide intervals reflect the tiny, selected subset and cannot establish equivalence or improvement. No inference errors were logged, so the null result is not attributable to failed model requests. The result is consistent with the two-step training budget being too small, with imperfect reward verification, or with the reward not helping navigation; this probe cannot distinguish those explanations.

Verifier failure analysis. Table 4 compares the frozen verifier with labels fixed before its predictions were obtained. Seven of eight reviewed completions receive Y, but only 7 of 13 Y predictions match a reviewed completion; two are on clear non-completions and four on ambiguous turns. One false positive occurs after the agent has already entered the requested room, exposing a failure to distinguish current state from an event newly completed this turn. Motion gates alone do not eliminate this error. These selected examples cannot establish deployment precision, but they make verifier reliability a concrete limitation of the present reward.

6. Discussion and Limitations

The main risk is reward misspecification. A VLM may mistake a visually similar doorway for the requested one, and hard displacement thresholds may miss legitimate small movements. Ordered one-at-a-time veri-

Table 4. Blind review of 49 selected val-unseen event transitions from 12 episodes. Rows are review labels and columns are frozen verifier predictions. U means evidence insufficient; the review was AI-assisted and has no independent human adjudication.

	Pred. Y	Pred. N	Pred. U
Review Y	7	0	1
Review N	2	15	12
Review U	4	1	7

Table 5. Exploratory evaluation on the same 16 R2R val-unseen episodes from 11 scenes. Mean goal distance is measured in meters. One generation per episode; the differences are too small for a performance claim.

Checkpoint	SR	SPL	Goal dist.
SFT initialization	6/16	0.3441	7.784
Destination only	6/16	0.3521	7.030
EventTrace	6/16	0.3529	6.821

fication can also miss rapid completion of two events in one turn. The verifier receives before/after frames rather than a continuous video; event timestamps are only turn precise. Finally, a trajectory-level GRPO advantage offers limited credit assignment to the action that completed the event.

A full comparison requires human-labeled and independently adjudicated verifier examples, including wrong-door and see-without-crossing cases, plus a separate parser audit. Training should use a substantially larger R2R train sample and multiple seeds, with equal compute for destination-only and event rewards. The final test must use the complete val-unseen split, keep episode IDs and decoding fixed across checkpoints, and report SR, SPL, navigation error, event precision/recall, and latency. Ablations of event ordering, abstention, motion gates, reward budget, and turn-level advantage would isolate why the reward helps or fails. Until those tests are completed, EventTrace is a tested method prototype and a paper draft, not a submission-ready empirical claim.

7. Conclusion

We formulated a path-free semantic process reward for closed-loop VLN that verifies ordered instruction events against executed motion and before/after views. In a controlled R2R training pilot, the reward enters the online GRPO loop, confirms intermediate events, and preserves destination priority. A 16-episode val-unseen probe finds the same 6/16 SR for SFT, destination-only, and event checkpoints. A sepa-

rate blind review finds substantial verifier disagreement on selected transitions. Larger training, full-split evaluation, and human-adjudicated event labels remain necessary to determine whether semantic events improve navigation.

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